Research Report

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Introduction

Crime and offences remain major global challenges in modern society. Governments, researchers and international organizations analyse crime statistics in order to improve public safety and understand social problems.

This report examines selected categories of global offences and presents simplified example statistics related to crime trends in different world regions. The analysis was conducted using R and Quarto.

Types of offences

Offences are illegal acts that violate the law. Different countries classify offences differently depending on their legal systems.

The main categories discussed in this report include: - property crimes - violent crimes - cybercrime - financial crimes - drug-related offences

Property crimes include theft, burglary and vandalism, while violent crimes include assault, homicide and domestic violence.

Cybercrime has become increasingly important because of technological development and growing internet usage.

Example dataset

The following dataset presents simplified example data showing estimated offence cases in different world regions.

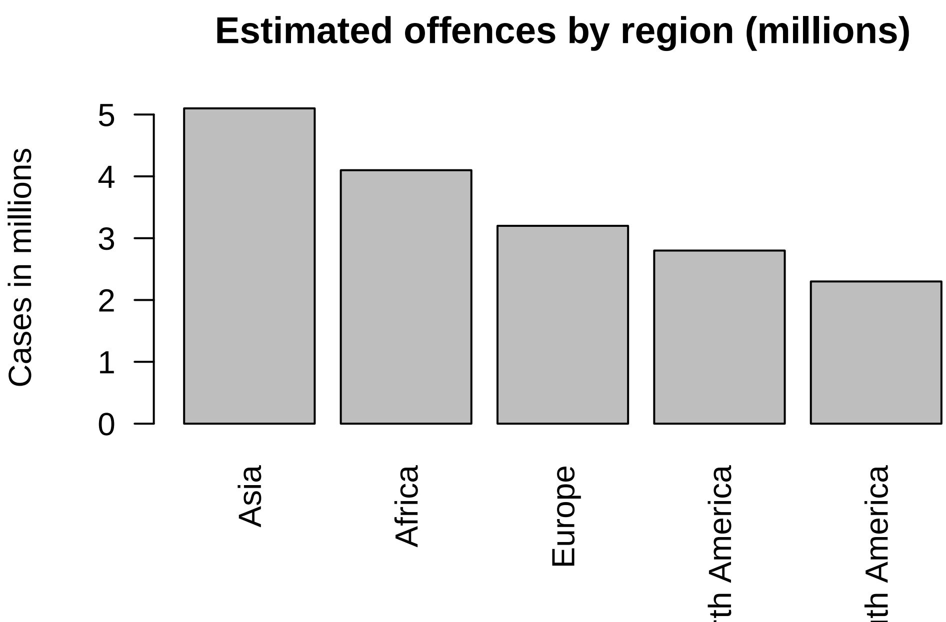
```r
global_data <- data.frame(
  Region = c("Asia", "Africa", "Europe", "North America", "South America"),
  Cases_Millions = c(5.1, 4.1, 3.2, 2.8, 2.3)
)

global_data
```

```
         Region Cases_Millions
1          Asia            5.1
2        Africa            4.1
3        Europe            3.2
4 North America            2.8
5 South America            2.3
```

Offences by region

The data can be visualised using a bar chart.

```r
barplot(
  global_data$Cases_Millions,
  names.arg = global_data$Region,
  main = "Estimated offences by region (millions)",
  ylab = "Cases in millions",
  las = 2
)
```



Statistical summary

```r
summary(global_data$Cases_Millions)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    2.3     2.8     3.2     3.5     4.1     5.1
```

```r
mean(global_data$Cases_Millions)
```

```
[1] 3.5
```

```r
max(global_data$Cases_Millions)
```

```
[1] 5.1
```

```r
min(global_data$Cases_Millions)
```

```
[1] 2.3
```

The average number of offences across the regions is presented above. Asia records the highest value, while South America records the lowest value in this example dataset. The graph suggests that Asia records the highest number of offences in this example dataset. One possible explanation is the large population size and high level of urbanisation. Africa and Europe also show relatively high numbers of cases, while South America presents the lowest value in this simplified example. # Types of offences worldwide

The next dataset presents selected offence categories.

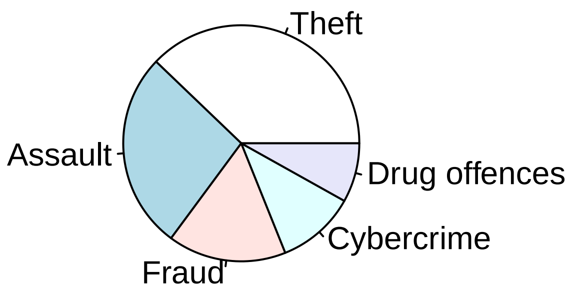
```
crime_types <- data.frame(
  Type = c("Theft", "Assault", "Fraud", "Cybercrime", "Drug offences"),
  Cases_Millions = c(7, 5, 3, 2, 1.5)
)

crime_types
```

```
           Type Cases_Millions
1         Theft            7.0
2       Assault            5.0
3         Fraud            3.0
4    Cybercrime            2.0
5 Drug offences            1.5
```

Comparison of offence categories barplot(crime_types$Cases_{M}illions, names.arg = crime_{t}ypes$Type, main = “Global offence categories”, ylab = “Cases in millions”, las = 2) The chart demonstrates that theft and assault remain among the most common offence categories worldwide.

```
pie(
  crime_types$Cases_Millions,
  labels = crime_types$Type,
  main = "Share of offence categories"
)
```

Share of offence categories



Cybercrime also continues to grow rapidly because of technological development and digital communication.

Factors influencing crime Several social and economic factors may influence crime levels, including: poverty unemployment lack of education inequality political instability

Urbanisation and population growth may also contribute to higher crime rates in some regions. Technology and crime Technology has both positive and negative effects on crime.

Positive effects include: improved surveillance systems digital evidence collection faster communication between law enforcement agencies

Negative effects include: cyberattacks online fraud identity theft financial scams

International cooperation Countries cooperate internationally to reduce crime and improve security. Organizations such as INTERPOL support information sharing and international investigations. International cooperation is especially important in fighting cybercrime and financial offences that cross national borders.

Limitations of the data The statistics used in this report are simplified illustrative examples. Real-world crime statistics may vary depending on: reporting systems legal definitions differences between countries underreporting of crimes

This means that actual crime levels may be higher than official data suggests.

Conclusion

Global offences remain a significant issue in the modern world. Different types of crime require different prevention strategies and international cooperation.

The analysis shows that: - crime affects all regions of the world - theft and assault remain common offences - cybercrime continues to increase - technology changes the nature of modern crime

Using R and Quarto makes it possible to analyse and present crime-related data in a clear and organised way.